BEDFORD BOROUGH WATER TREATMENT PLANT UPGRADES

Got Manganese?: Carmen Hagerty, Kate Hanson, Samantha Lee, Flannery McNair, Shria Shyam

12-401: CEE Design | Profs. Sarah Christian and Fethiye Ozis | Technical Advisor: Tyler McGraw | Fall 2024

We are grateful for the support of Professor Sarah Christian, Professor Fethiye Ozis, technical advisor Tyler McGraw (Stiffler McGraw), and John Whitmore (water treatment plant superintendent).

Scan for sources and more information.

PROBLEM DEFINITION

PROBLEM: Bedford Borough struggles with **high-manganese water**, uses **unsafe disinfection**, and has **non-compliant chemical storage**.

FUNCTION: Treat and supply potable water to Bedford Borough.

SUBFUNCTION 1: Produce effluent with **manganese** concentration < 0.05 mg/L.

Codes: Safe Drinking Water Act (USEPA 2015)

SUBFUNCTION 2: Provide a safe location for **chemical storage**.

Codes: National Fire Protection Association (NFPA)

SUBFUNCTION 3: Provide a means for **safe disinfection**.

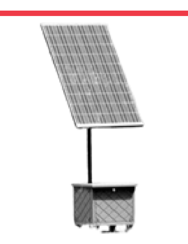
Codes: 4-log removal of viruses guidelines (PA DEP 2015)

CONSTRAINTS:

- The cost cannot exceed the **$3 million budget**.
- The treated water must **satisfy EPA regulations**.
- The design must account for **existing infrastructure and topography**.

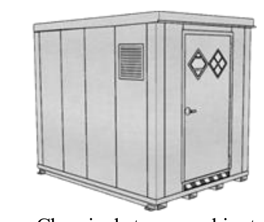
DESIGN DETAILS

SOLUTION: The following components are proposed to mitigate the problems at the Bedford Borough Water Treatment Plant.

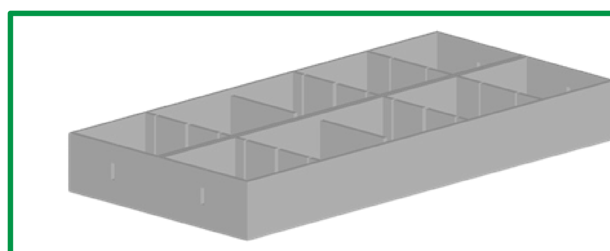

Solar-powered aerator (ProLake 2024).


Aerator (Living Water Aeration 2024).

Solar-powered bottom-up aerators are installed in the reservoir to prevent manganese mobilization. Dissolved oxygen sensors help optimize aerator operation.


Chemical storage cabinet (USA Safety 2024).

A prefabricated **chemical storage building** is installed at the raw water site. A new **carrier water system** transports chemicals from the new building to their delivery points.


Baffle design, this proposal.

Baffles are added to the reactor to improve disinfection efficiency. Chlorine gas is replaced with safer **sodium hypochlorite**.

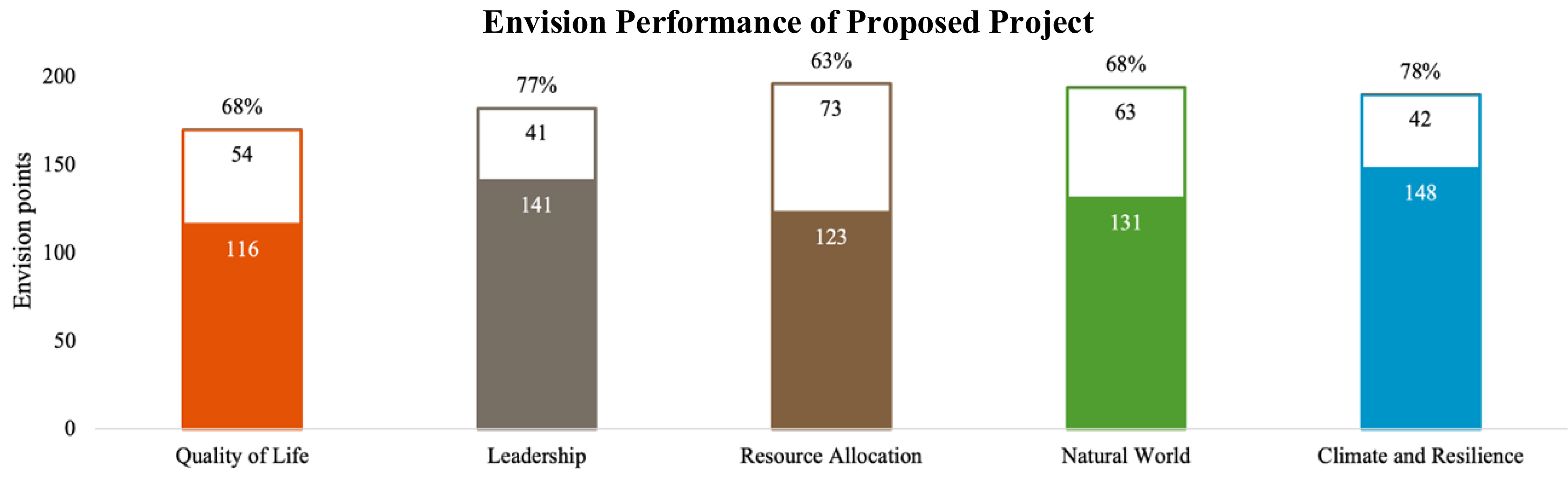
THE PROPOSED DESIGN MEETS ALL CONSTRAINTS.

- The 50-year net present value (NPV) is estimated as **$183,190**.
- Sufficient oxygen is provided to **immobilize metals below the EPA limit**.
- The design adds onto **existing infrastructure and topography**.

PERFORMANCE SUMMARY

OBJECTIVE 1: MAXIMIZE SUSTAINABILITY.

The **Envision framework** guides sustainable infrastructure projects.

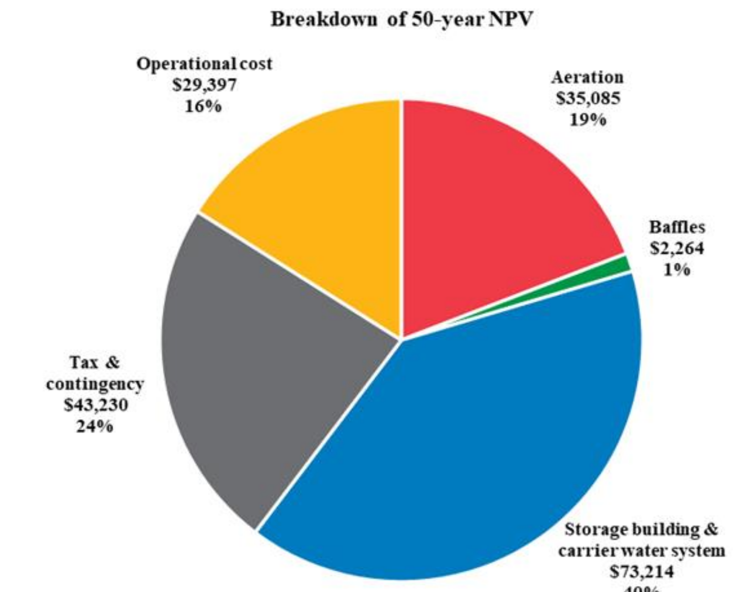


- **Lower chemical use** and **safer disinfection** improves performance in **quality of life**, **leadership**, **natural world**, and **climate and resilience**.
- **Solar panels** improve **resource allocation** through renewable energy.

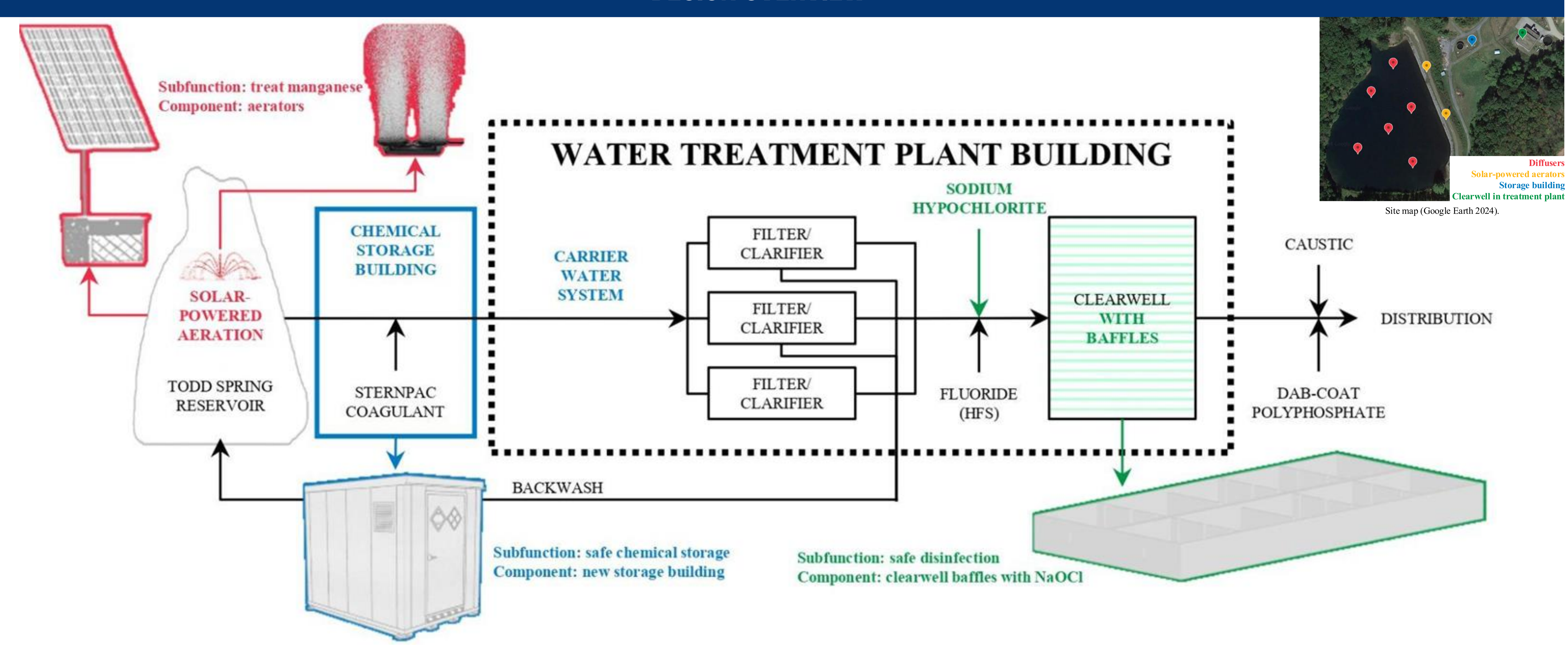
OBJECTIVE 2: MINIMIZE COST.

- **Baffles** increase disinfection efficiency, lowering annual chemical costs.
- **Solar panels** eliminate annual electricity costs for aeration.
- **Prefabrication** lowers construction costs for the chemical storage building.

Capital cost	$153,793
Capital cost includes aeration, baffles, storage building & carrier water system, and tax & contingency.	
Annual operational cost	$2,130
50-year NPV	$183,190



DESIGN OVERVIEW



Site map (Google Earth 2024).

FUTURE WORK

- **Emerging pollutants** may require new water treatment processes.
- **Future water treatment regulations** may necessitate additions.
- **Future maintenance** will keep systems updated and functional.

LESSONS LEARNED

- **Nature-based solutions** (e.g., aeration) can help decrease chemical use.
- **Solar panels** increase capital cost but decrease operational cost.
- Pipes can follow **natural slopes** to eliminate pumping requirements.